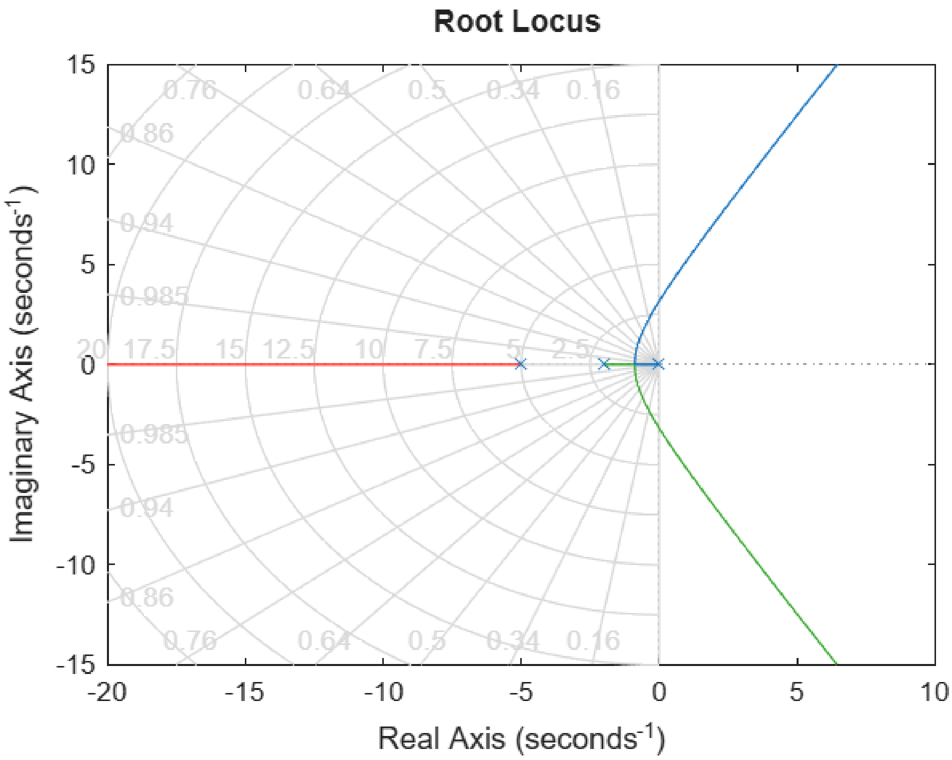
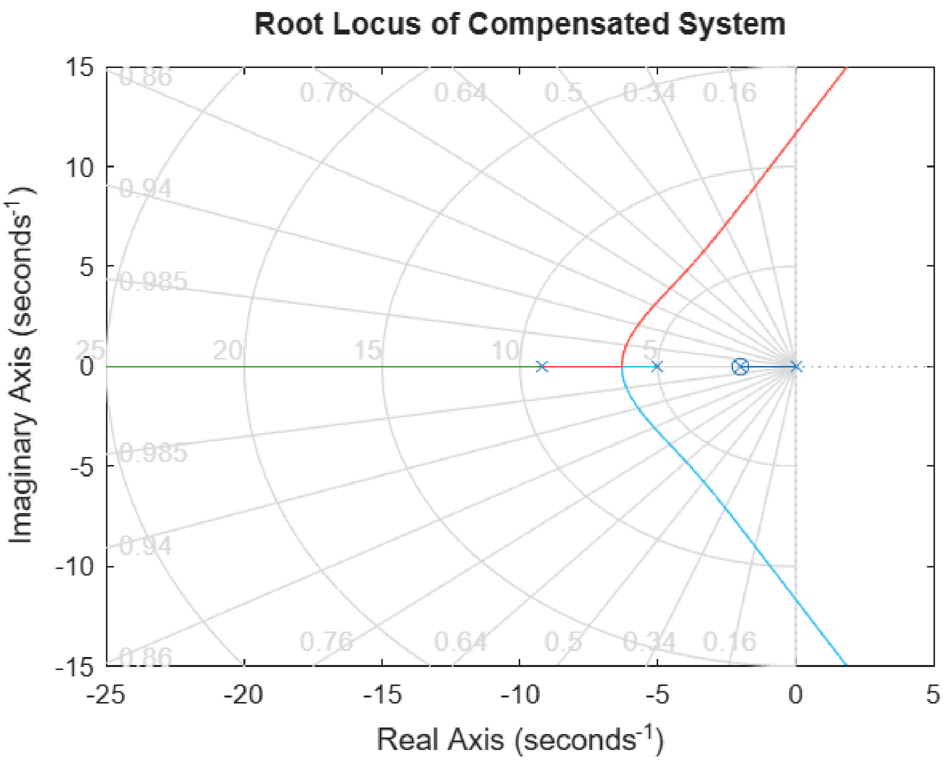
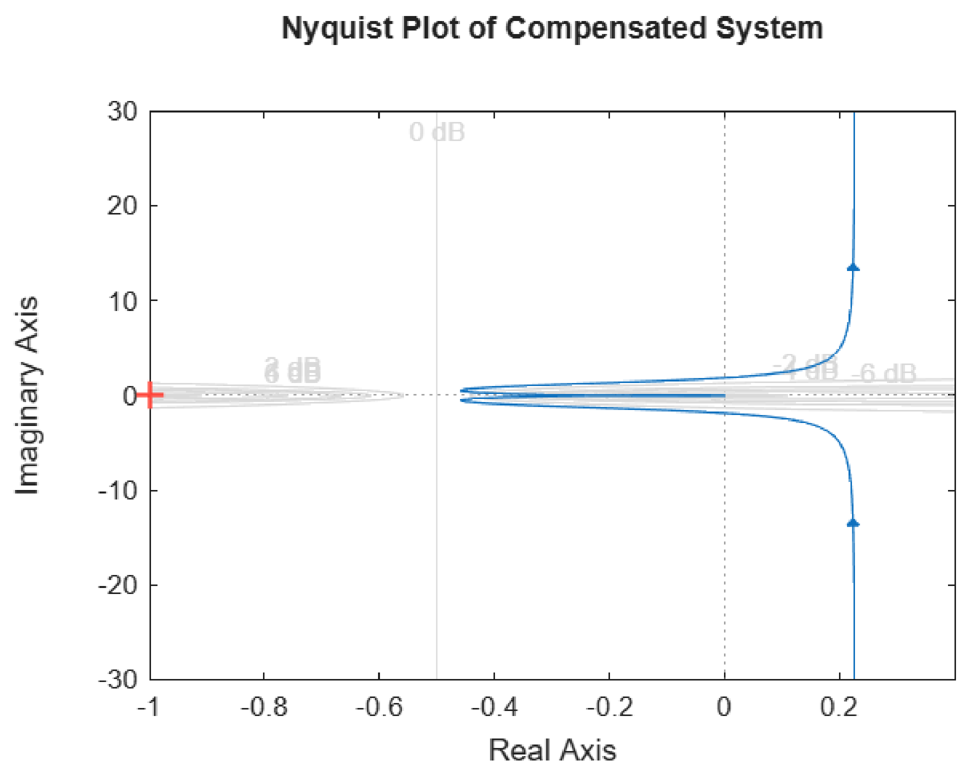
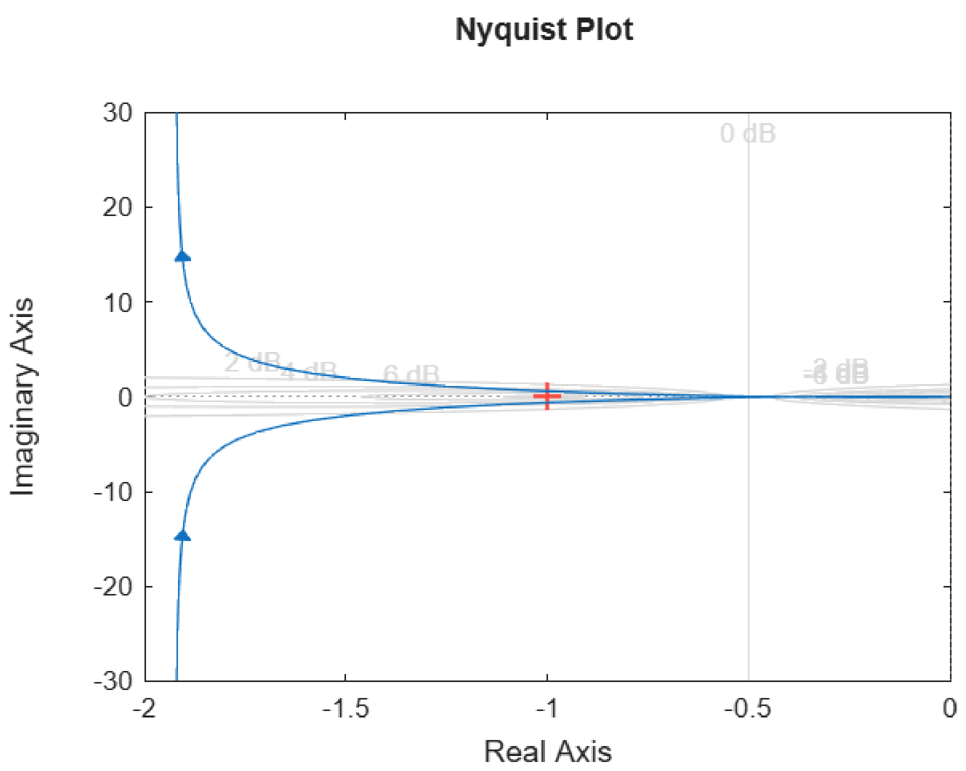
```
clear; clc; close all;
s = tf('s');
G = 1 / (s*(s+2)*(s+5));
alpha = 0.2175;
T = 0.5;
K_gain = 27.5;
Gc_single = (1 + T*s) / (1 + alpha*T*s);

Gc_total = K_gain * (Gc_single)^2;

G_open_comp = Gc_total * G;
G_cl_comp = feedback(G_open_comp, 1);
G=K_gain*G;
figure(1);
rlocus(G);
title('Root Locus ');
grid on;

figure(2);
rlocus(G_open_comp);
title('Root Locus of Compensated System');
grid on;

figure(3);
nyquist(G_open_comp);
title('Nyquist Plot of Compensated System');
grid on;
```

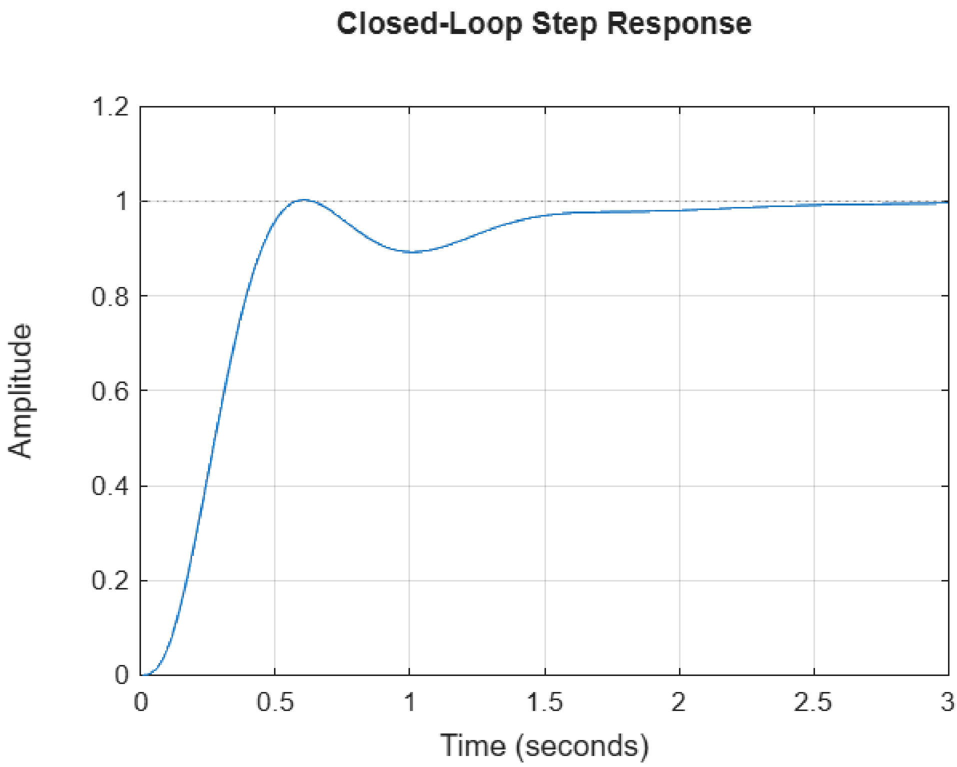
```
figure(4);
nyquist(G);
title('Nyquist Plot ');
grid on;

figure(5);
step(G_cl_comp);
title('Closed-Loop Step Response');
grid on;

S = stepinfo(G_cl_comp);
fprintf('--- Performance Results ---\n');
fprintf('Peak Overshoot: %.2f%%\n', S.Overshoot);
fprintf('Settling Time (2%%): %.2f s\n', S.SettlingTime);

[Gm, Pm, Wcg, Wcp] = margin(G_open_comp);
fprintf('Phase Margin: %.2f degrees\n', Pm);
fprintf('Gain Margin: %.2f dB\n', 20*log10(Gm));

figure(6);
margin(G_open_comp)
```


Root Locus
Imaginary Axis (seconds⁻¹)
Real Axis (seconds⁻¹)
15
10
5
0
-5
-10
-15
-20
-15
-10
-5
0
5
10
0.76
0.64
0.5
0.34
0.16
0.86
0.94
0.985
20
17.5
15
12.5
10
7.5
5
2.5


Root Locus of Compensated System
Imaginary Axis (seconds⁻¹)
Real Axis (seconds⁻¹)
15
10
5
0
-5
-10
-15
-25
-20
-15
-10
-5
0
5
0.86
0.76
0.64
0.5
0.34
0.16
0.94
0.985
25
20
15
10
5


Nyquist Plot of Compensated System
0 dB
2 dB
6 dB
-2 dB
-4 dB
-6 dB
30
20
10
0
-10
-20
-30
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
Imaginary Axis
Real Axis


Nyquist Plot
0 dB
2 dB
4 dB
6 dB
-2 dB
-6 dB
30
20
10
0
-10
-20
-30
-2
-1.5
-1
-0.5
0
Imaginary Axis
Real Axis




```
--- Performance Results ---
Peak Overshoot: 0.26%
Settling Time (2%): 1.98 s
Phase Margin: 67.28 degrees
Gain Margin: 13.52 dB
```